TRƯỜNG ĐẠI HỌC FPT

CAPSTONE PROJECT REGISTER

Class: **Duration time**: from 09/2026 To 03/2027

(*) Profession: Software Engineer **Specialty**: SE [X]

(*) Kinds of person make registers: Lecturer [x] Students

1. Register information for supervisor (if have)

No.	Full name	Phone	**E-Mail**	**Title**
Supervisor	Nguyễn Ngọc Lâm		lamnn15@fpt.edu.vn	

2. Register information for students (if have)

	Full name	**Student code**	**Phone**	**E-mail**	**Role in Group**
1	Nguyễn Bỉnh Khiêm	SE182089	(removed)	KhiemNBSE182089@fpt.edu.vn	Leader
2	Phan Tấn Phát	SE182038	(removed)	PhatPTSE182038@fpt.edu.vn	Member
3	Ngô Minh Tân	SE182434	(removed)	TanNMSE182434@fpt.edu.vn	Member
4	Đinh Hoàng Phúc	SE182107	(removed)	PhucDHSE182107@fpt.edu.vn	Member
5	Nguyễn Kim Thiên Ngân	SA180013	(removed)	NganNKTSA180013@fpt.edu.vn	Member

3. Register content of Capstone Project

(*) 3.1. Capstone Project name:

English: Hotel Booking and Personalized Recommendation Platform

Vietnamese: Nền tảng đặt phòng khách sạn và gợi ý cá nhân hóa

a. Context:

The Vietnamese hospitality industry has grown rapidly in recent years, with hotels of all sizes competing for online bookings across distribution channels such as Booking.com, Agoda, and direct websites. While large hotel chains can afford specialized digital systems, small and mid-sized hotels often rely on manual processes for guest interaction, social media marketing, and customer care, and struggle to reach travelers through a trusted booking channel of their own.

From the guest perspective, the booking journey is fragmented: travelers compare prices on multiple platforms, send messages through different channels, and frequently wait hours for a response about room availability, while post-stay follow-up is rarely personalized. From the management perspective, independent hotels find it costly to maintain a consistent online presence, produce engaging content regularly, and respond promptly to comments, reviews, and direct messages.

StayReco addresses these gaps as a multi-tenant booking and customer-engagement platform that connects many small and mid-sized hotels (hotel partners) with travelers in a single place. Hotel partners onboard by uploading a business license and submitting a verification request, which a platform manager reviews and approves; each partner then manages its own rooms, pricing, staff, and analytics within an isolated tenant workspace, while a platform operator oversees partners, commissions, and policy compliance. Travelers browse and book across partner hotels with the help of an AI assistant, which also supports each partner's customer care and marketing activities.

Functional requirement :

– Common (Authorized User)

● Log out of the system.

● Manage personal profile information (name, password, avatar, contact details).

● Receive system notifications.

– Guest (anonymous visitor)

● Register an account, log in, and recover a forgotten password.

● View the hotel blog.

● Browse and search rooms and view detailed room information.

● Chat with the AI assistant to get room recommendations based on the current session and search behavior.

– Customer (registered guest)

● Make a room booking, which includes checking room availability and making payment.

● Manage bookings, including modifying or cancelling a reservation.

● View booking history and upcoming bookings.

● Receive personalized hotel and room recommendations based on preferences, search behavior, and booking history.

● Upload review photos after a stay.

● Redeem discount vouchers and earn loyalty points.

– Hotel Staff

● Confirm bookings and view booking information.

● Manage the check-in process (including assigning rooms) and the check-out process.

● Monitor live AI conversations and take over a chat when escalation is required.

● Respond to guest inquiries and complaints, viewing the related booking information.

● Process refunds, including verifying the booking policy, and handle refund requests.

● Receive notifications alerts triggered by unusual booking patterns or negative reviews.

● Review AI-generated marketing content, then edit and approve it before publication, and schedule content publication, including connecting supported social media accounts.

● Monitor marketing content performance, including reach, engagement, and conversion metrics.

– Hotel Partner (hotel owner / operator)

● Upload a business license, submit a verification request, and view verification status.
● Manage hotel information, hotel images, and hotel policies.
● Manage room inventory, including adding new rooms, managing availability, and updating room pricing and information.
● Configure rule-based room pricing, including seasonal base-rate rules.
● Manage the hotel's loyalty points and voucher program, including configuring point-earning rules, issuing discount vouchers, and setting redemption conditions.
● View the analytics dashboard, including hotel revenue, occupancy rate, booking sources, and customer satisfaction trends.
● Manage staff accounts and assign roles and permissions.
● Configure AI content-generation settings (brand voice, tone, and content templates) for their own hotel.
● Manage promotional campaigns and discount codes applied on top of the base room price (final price = base price - applicable discount).
– Admin (platform administrator)
● Manage the system dashboard and view system statistics.
● Manage user accounts, including approving, activating, and deactivating accounts.
● Manage payment configurations, including verifying payment status and viewing transaction logs.
● Monitor system logs, including audit trails and security events.
● Configure system-level AI settings (model selection, API keys, global usage limits, and safety filters).
– Platform Manager
● Approve hotel partner verification requests.
● Monitor hotel partners, including their booking activities and platform performance.
● Detect policy violations and request partner suspension when necessary.
● View platform revenue, including commission reports and platform analytics.
● View system growth metrics.

Non-functional requirement:

● Performance: respond to user requests in a timely manner appropriate to the complexity of each operation, and support concurrent usage suitable for a multi-tenant environment.
● Security: use JWT-based authentication and role-based authorization; hash passwords; encrypt sensitive data at rest and in transit; process payments through secure third-party gateways without storing raw card credentials.
● Multi-tenancy & Data Isolation: isolate each hotel partner's data at the tenant level so a partner can access only its own hotel's rooms, bookings, staff, and analytics, enforced through tenant-scoped access control.
● Usability: provide a responsive, intuitive interface on desktop and mobile, allowing guests to search, book, and pay within a simple, clear workflow.
● Reliability: handle booking, payment, and check-in/check-out operations without data inconsistency and display meaningful error messages when failures occur.
● Scalability: support future expansion in the number of hotel partners, users, bookings, and AI services.

● Maintainability: follow modular design, clean architecture, and consistent naming conventions, and remain easy to test, maintain, and extend.
● Compatibility: support common browsers (Chrome, Edge, Safari, Firefox) and remain accessible on desktop and mobile through responsive design.
● API Standards: follow RESTful conventions with consistent status codes and response formats.
● Data Privacy: protect personal data according to Vietnamese regulations (Decree 13/2023/ND-CP) and GDPR principles, allowing users to access only data permitted by their role.

(*) 3.2. Main proposal content (including result and product)

Proposed Solutions:

StayReco is a multi-tenant platform that centralizes hotel booking and customer engagement for small and mid-sized hotels, connecting hotel partners and travelers in a single system. Key solutions include:

● Unified Booking Marketplace: Lets guests search and book rooms across multiple partner hotels, check availability, complete payments, and receive QR-code e-vouchers in one platform.
● Conversational AI Assistant: An LLM-powered assistant that answers room-related questions and gives personalized recommendations, supporting guests in Vietnamese or English.
● Hotel Partner Onboarding & Verification: Hotel partners register, upload a business license, and submit a verification request that a platform manager approves before the hotel goes live, with a tenant-isolated workspace for each partner.
● Personalized Recommender System: Analyzes guest preferences, search behavior, and booking history to surface relevant hotel and room recommendations, helping travelers discover suitable options and improving booking conversion rates; recommendations surface both inside the AI chat and on browse and discovery pages, with the chatbot calling the recommender service rather than duplicating it.
● Customer Care Automation: Provides automated replies for common questions and a unified inbox for staff to manage guest messages, reviews, and complaints, with human handoff when needed.
● AI-Assisted Marketing Support: Generates draft captions, blog content, and promotional ideas for hotel partners to review, edit, and approve before scheduled publication.
● Partner Analytics & Platform Operations: Hotel partners monitor their own bookings, revenue, occupancy, and satisfaction, while a platform manager oversees partners, commission and platform-revenue reports, and policy compliance.

Applied Theory:

● Students should apply the software development process and UML 2.0 in modeling the system.
● The documents include User Requirements, Software Requirement Specification, Architecture Design, Detail Design, System Implementation and Testing Document, Installation Guide, source code, and deployable software packages.

– Framework: Express with RESTful API
– AI Service: Python (FastAPI), OpenAI
– Database: PostgreSQL as the primary database
– Multi-tenancy: tenant-scoped data isolation (e.g., X-Tenant-Id header with row-level tenant filtering)
– Caching / Session / Queue: Redis (when needed)
– AI Integration: OpenAI API for chatbot and content draft generation

– Vector Search: PostgreSQL pgvector extension for RAG-based knowledge retrieval (FAQ, hotel policies) and embedding-based room recommendations (content-based similarity combined with behavioral signals from search and booking history)
– Hosting BE: Docker + cloud hosting (VPS / managed platform)
– Web Client: TypeScript, React.js, TailwindCSS (responsive design for desktop and mobile browsers)
– Mobile Client: React Native with Expo, TypeScript, NativeWind — cross-platform iOS & Android guest application
● **External integrations:**
– Payment Gateway: VNPay or SePay, selected based on integration availability
– Media & document storage: Cloudinary (or object storage) for hotel images and business-license documents via Multer uploads
– Notifications: Firebase Cloud Messaging for push notifications
– Email service: SendGrid/Nodemailer or equivalent for transactional emails

Products (Expected Deliverables):

● Web API for the entire system (Express RESTful API + AI service layer) with multi-tenant data isolation.
● Cross-platform mobile application (React Native / Expo) for Guest and Customer users on iOS and Android — AI chatbot, QR voucher, push notifications, and loyalty wallet.
● Responsive web application for Guest, Customer, and Hotel Staff (accessible on both desktop and mobile browsers).
● Hotel Partner portal for onboarding and verification, hotel and room management, pricing, staff, and partner analytics.
● Platform operations console for the Platform Manager and Admin — partner monitoring, verification approval, commission and platform-revenue reports, and policy enforcement.
● AI service layer for chatbot, content draft generation, and personalized recommendations.
● Database system for partners, users, rooms, bookings, payments, messages, reviews, and promotions.
● Documentation package including SRS, design documents, testing documents, installation guide, and user manual.
● Source code repository and deployable software package.

Proposed Tasks:

● Task package 1: Develop the backend system and database — design models for partners, users, rooms, bookings, payments, reviews, messages, and promotions; build APIs for authentication, hotel-partner onboarding and verification, room and booking management, payment processing, and notifications; implement multi-tenant data isolation, role-based access control for all roles, and booking-commission tracking for platform revenue.
● Task package 2: Develop the web application — build role-based interfaces for Guest, Customer, Hotel Staff, Hotel Partner, Admin, and Platform Manager, including room search and booking, customer-care inbox, marketing content studio, partner onboarding and management, analytics dashboards, and the platform operations console, all connected to backend APIs.
● Task package 3: Develop the AI assistant and customer-care services — build the conversational AI chatbot that answers guest questions and provides support, implement AI-assisted responses, support human handoff for complex cases, and build the personalized

recommender service that suggests hotels and rooms from guest preferences, search behavior, and booking history.

● Task package 4: Develop the AI marketing support module — build content generation for captions, blog posts, and promotions, with hotel staff review and approval, and support scheduling or manual publishing.

● Task package 5: Build, deploy, and test the entire system — integrate backend, web, and AI services; conduct unit, integration, system, and user acceptance testing; deploy to a cloud or hosting environment.

● Task package 6: Prepare required documents — system analysis and design, test plan and testing report, installation manual, and user manuals for Guest, Customer, Hotel Staff, Hotel Partner, Admin, and Platform Manager.

3.3. Research Information

a. Research Problem / Research Question:

Multi-tenant hotel booking platforms face a dual cold-start problem: new hotel partners have no booking history, and most guests are anonymous or first-time visitors with sparse data. Content-based recommenders ignore behavior; pure behavioral models fail when data is sparse. Research question: How can a hybrid, density-adaptive recommender — fusing content-based hotel/room embeddings with behavioral signals, weighted by interaction density — improve recommendation quality under cold-start conditions while preserving tenant data isolation?

b. Research Objectives:

1. Characterize cold-start patterns for both new hotel partners and new guests in a multi-tenant setting.
2. Design a density-adaptive hybrid recommender that routes between content-dominant and behavior-dominant inference based on available interaction density.
3. Build the recommender as a shared service used by both the AI chatbot and browse/discovery pages.
4. Evaluate against content-only and behavior-only baselines (Precision@K, Recall@K, NDCG, cold-start coverage).

c. Research Scope & Methodology:

Scope : Recommender microservice within StayReco covers hotel/room recommendation

Methodology: Literature review of hybrid and density-adaptive recommenders; data from guest search/booking logs and hotel/room metadata (supplemented with synthetic data for cold-start stress-testing); pgvector-based content embeddings + behavioral signal encoding; density-adaptive fusion mechanism adapted from trust/density-based routing in social recommendation; implemented in Python (FastAPI) +DBVector

d. Expected Scientific Contribution:

1. Adaptation of density-adaptive fusion (from social recommendation) to dual-sided cold-start in multi-tenant hotel booking.
2. Reusable hybrid recommender architecture generalizable to other multi-tenant marketplaces.
3. Empirical evidence that density-adaptive fusion outperforms single-signal baselines under cold-start.

e. Related Works / Literature Review (Preliminary):

1. Hybrid recommender systems: combining content-based and collaborative filtering to offset each method's weaknesses (Burke, 2002); weighted, switching, and cascade hybridization strategies as foundations for the proposed content–behavior fusion.
2. Cold-start recommendation: content-based fallback (Lika et al., 2014) and cross-domain/meta-learning approaches (Vartak et al., 2017) for new users/items with no interaction history — directly relevant to the dual-sided (partner + guest) cold-start problem here.
3. Graph-based recommenders: NGCF (Wang et al., 2019) and LightGCN (He et al., 2020) for modeling high-order collaborative signals via user–item graphs, relevant to the behavior-dominant branch.
4. Other comments (propose all relative things if have).

Supervisor (If have) *(Sign and full name)*	HCM, date 02/07/2026 **On behalf of Registers** *(Sign and full name)*

SEP490 · CAPSTONE PROJECT · REVIEW 2 · BM03

Registration Form – Record of Changes

1. Project Information

Project code	FA26SE056
Project name	StayReco – Hotel Booking and Personalized Recommendation Platform
Team	GFA26SE71 – Nguyen Binh Khiem (leader), Dinh Hoang Phuc, Nguyen Kim Thien Ngan, Ngo Minh Tan, Phan Tan Phat
Supervisor	Nguyen Ngoc Lam
Current registration version	1.0 (signed 02/07/2026)
Proposed version	1.0 – no changes
Version	1.0
Date	02/11/2026

2. Changes

☒ No changes to the current registration form

No.	Section	Current content	Proposed content	Reason	Proposed by
1	—	No changes	—	The supervisor reviewed the current form and confirmed that it stays as approved.	Nguyen Binh Khiem

3. Decision at Review 2 (completed by reviewers)

No.	Decision (Accepted / Adjusted / Rejected)	Reviewer note
1		